Pakistan Journal of Engineering and Technology
ISSN (Print): 2664-2042; ISSN (Online): 2664-2050
Received: 2025-02-28
Accepted: 2025-04-16
https://doi.org/10.51846/vol8iss1pp21-31

PAKJET
Pakistan Journal of Engineering & Technology

Article

Consensus-Based Autonomous Microgrid Secondary Frequency Control with Communication Impairments

Sunday Akinwale Olayanju [1,*], Folasade Mojisola Dahunsi [2], and Akinlolu Adediran Ponnle [3]

[1] Department of Electrical and Electronics Engineering, University of Ilorin, Ilorin, Nigeria
[2] Department of Computer Engineering, The Federal University of Technology, Akure, Nigeria
[3] Department of Electrical and Electronics Engineering, The Federal University of Technology, Akure, Nigeria
* Correspondence: Sunday Akinwale Olayanju (olayanju.sa@unilorin.edu.ng)

Abstract: The frequency stability requirement of islanded microgrids is dependent on the control of electronic inverters; therefore, it is necessary to use proper control methods to improve the robustness of the microgrid system in the presence of impairments. While primary droop-based control methods can keep the frequency deviation within allowable limits, they are weak in synchronizing frequencies of distributed generators to the reference value. In this work, a consensus-based approach is employed for secondary frequency control in autonomous microgrids to ensure frequency synchronization among the distributed generators (DGs) and to mitigate deviation in frequency of operation. In the approach, the model of the microgrid system was used to develop a consensus-based secondary control scheme, and tracked errors in frequency and local active power with information received over impairment-corrupted communication links among the DG units were considered. The performance of the proposed method was evaluated using a simulation of a four-DG microgrid test system implemented in MATLAB/SimPowerSystem software environment. The simulation considered load perturbation among the DGs, communication noise, link failure, and link time delay. The results from the simulation showed that the method achieves frequency synchronization with load perturbations despite the presence of communication impairments within the communication links of the DGs.

Keywords: Consensus, Microgrid, Noise, Delay, Link Failure, Frequency Control

1. Introduction

1.1. Background and Motivation

Microgrids (MGs) are then thought of as electric power networks with low voltages and usually small-scale capacities. The key components of a microgrid include distributed energy resources (DERs) such as solar panels, wind turbines, fuel cells, and energy storage systems. In addition to these, there are other important components such as Intelligent Electronic Devices (IEDs) which include Internet of Things (IoT) sensors, smart meters, and energy management software or platforms. Real time monitoring and control of energy generation, distribution, consumption and storage is also used to build microgrids [1], [2]. A microgrid can operate in either grid-tied or autonomous mode. DERs typically serve as the source of electric energy for distributed generators (DGs), whether the microgrid is operating autonomously or supplying power to the main grid [3], [4]. The hierarchical control structure is the fundamental approach used for autonomous microgrid control. The structure of this is three level primary, secondary, tertiary control. Firstly, primary control is based on droop control for the objective of keeping voltage and frequency stable, as well as for the sharing of the active and reactive power between DGs [5], [6].

With large line impedances, droop control is found to be weak as generally, deviations from voltage and frequency reference values and also inaccurate reactive power sharing occur. An abrupt drop in system frequency due to load variation may also be observed [5], [7], [8]. Secondary control is introduced to address or mitigate the limitations of droop-based primary control. Unlike primary control, secondary control requires communication among the DGs in the microgrid network [5], [9], [10]. When secondary control employs a centralized control structure, it is prone to a single point of failure, as information from all the DGs is aggregated at one centralized controller. A fault in this controller can lead to the collapse of the entire microgrid system. Additionally, heavy communication and computational burdens, along with poor scalability and reliability, are further disadvantages of using a centralized control structure [11], [12]. However, when secondary control is implemented using a distributed control structure, the issue of a single point of failure is effectively eliminated. Moreover, the need for an expensive communication network can be reduced by employing distributed secondary control structures based on multi-agent theory [13]. However, communication links are stochastic and stochastic disturbances make inaccurate measurement of states of the neighbor available to each DGs. Noisy measurement data can adversely impact sensitive electronic devices and lead to instability in the MG network [5], [14]. Tertiary control primarily handles energy management between the microgrid and the utility system, with optimal system performance as its

main objective [7]. The emergence of smart MGs has transformed the way electric energy is managed. Communication networks are essential for integrating renewable energy sources, energy storage, and energy management systems to ensure smooth system operation [15], [16]. Similar to this, information regarding system frequency and voltage restoration as well as synchronization of the DGs is also arranged to be communicated over the communication networks. MG control is by far the most important role of MG, namely ensuring that MG's voltage and frequency are within the reference values and sharing the active and reactive power of all DGs [17]. However, the complexity and type of communication networks employed pose significant challenges to the reliable operation and control of the smart MG system. Some of the major communication impairments in MG operation include noise and disturbances, communication link delays, link breakages and data dropouts, scalability issues, system complexity, and cybersecurity threats [13], [18].

1.2. Literature Review

With respect to the mitigation of communication impairments in islanded microgrids, several works in the literature have been proposed. A distributed frequency controller can be used to restore both the secondary voltage and frequency in sparse communication networks to tackle communication delays [1]. Distributed cooperative secondary control for microgrid management was implemented in [19] for voltage and frequency restoration. For non-uniform time delays in communication networks, distributed cooperative secondary frequency control has been applied with satisfactory results [17]. The burden of communication costs has been reduced through the implementation of intermittent information sharing among communicating DGs. The establishment of this scheme in sparse communication networks was achieved through iterative learning mechanisms [20]. The method in [5] addresses noisy communication links among DGs while tackling voltage and frequency synchronization in microgrids. A noise-resilient secondary control method was used, which considers generalized white Gaussian noise in communication links. In [11], a linear extended state observer was used to estimate model dynamics, where nonlinear models were converted to quasi-linear ones. The proportional distributed control law employed ensures frequency restoration and power sharing among the DGs. The active disturbance rejection control method improves the disturbance rejection performance of DGs' inverters. A resilient control algorithm for voltage and frequency restoration in microgrids, as used in [9], addresses state-dependent noise and communication time delays. The algorithm ensures active power sharing. To address random packet loss in communication networks, an output feedback controller based on distributed cooperative secondary control was employed for frequency and voltage restoration [21]. In [22], an optimal distributed hybrid model predictive secondary control method was implemented, using User Datagram Protocol (UDP) at the transport layer. The proposed method applies hybrid model predictive secondary control under random communication disorders. In this paper, attention is focused on distributed secondary frequency control using consensus-based control techniques under communication impairments in a multi-agent system. In [17], [20], the microgrid system is modeled as a multi-agent system, with DGs representing the agents. The problem of voltage and frequency regulation and synchronization control is transformed into a consensus problem in a multi-agent system, with a reference value pinned to at least one agent in the network. Reference [23] proposed a distributed control strategy based on a consensus algorithm. An event-driven control method is adopted to reduce communication burdens, and a ratio-consensus algorithm is utilized for frequency recovery under communication delays. The noise-resilient distributed secondary control method in [14] uses consensus protocols for voltage and frequency restoration. Intrinsic communication link delays among communicating DGs were taken into consideration. In [24], sliding mode control and consensus control was merged for achieving better stability, and virtual leader-following consensus control was developed to achieve voltage and frequency management. Consideration was also given to packet loss between communicating DGs. To coordinate DGs in islanded microgrids, [7] implemented a consensus-based distributed regulator for voltage and frequency, ensuring that the frequencies of all generators converge to a reference value..

1.3. Contribution

Previous works in the literature have study secondary frequency control when communication impairments impact the entire microgrid system. A lot of solutions have been provided to mitigate the effects of communication disturbance while DGs share control information. The importance of this study is based on the implementation of consensus based secondary frequency control using frequency error tracking algorithm. The approach in this work considers communication impairments when only one of the DG is affected. This approach clarified the effects of impairments on system operation and measures the ability of the secondary controller in synchronizing DG sharing information under impairments with the rest of the system.

2. Materials and Methods

2.1. Fundamentals of Graph Theory

For multi-agent topology representation, directed (digraphs) or undirected graphs can be utilized to enable distributed control across distributed generating sources (agents). It explains how an agent network's topology is organized. Given that $\boldsymbol{\mathcal{G}} = (\boldsymbol{\mathcal{V}}, \boldsymbol{\mathcal{E}}, \boldsymbol{\mathcal{A}})$ is a weighted digraph of order n, it consists of the set of edges $\boldsymbol{\mathcal{E}} \subseteq \boldsymbol{\mathcal{V}} \times \boldsymbol{\mathcal{V}}$, the set of nodes $\boldsymbol{\mathcal{V}} = \{\boldsymbol{\mathcal{V}_1}, \ldots \boldsymbol{\mathcal{V}_n}\}$, and a weighted adjacency matrix $\boldsymbol{\mathcal{A}} = [\boldsymbol{a_{ij}}]$, in which the adjacency elements $\boldsymbol{a_{ij}}$ are non-negative. Every agent in a network is represented by one or more digraph nodes. The communication links between linked nodes are indicated by the edges. Agent $\boldsymbol{i}$ is implied to acquire information from agent j by each edge $\boldsymbol{\mathcal{E}_{ij}} = (\boldsymbol{\mathcal{V}_i}, \boldsymbol{\mathcal{V}_j})$ [25], [26]. An agent using a

digraph only gets information from its neighbors $\boldsymbol{N_i}$. The weight of edge $\boldsymbol{\mathcal{E}_{ij}} = (\boldsymbol{\mathcal{V}_i}, \boldsymbol{\mathcal{V}_j})$ is represented by $\boldsymbol{a_{ij}}$, which equals 1 if $(\boldsymbol{V_i}, \boldsymbol{V_j})\ \boldsymbol{\epsilon}\ \boldsymbol{\mathcal{E}}$ and 0 otherwise. The set of node $\boldsymbol{\mathcal{V}_i}$ neighbors is indicated by:

$$\boldsymbol{N_i} = \{\boldsymbol{\mathcal{V}_j} \boldsymbol{\epsilon} \boldsymbol{\mathcal{V}}: (\boldsymbol{\mathcal{V}_i}, \boldsymbol{\mathcal{V}_j})\ \boldsymbol{\epsilon}\ \boldsymbol{\mathcal{E}}\} \tag{1}$$

$\boldsymbol{\mathcal{D}} = \boldsymbol{diag}\{\boldsymbol{d_i}\} \in \boldsymbol{R^{N\times N}}$ is the definition of the in-degree matrix, where $\boldsymbol{d_i} = \sum_{j\in N_i} \boldsymbol{a_{ij}}$. The formula for the Laplacian matrix is $\boldsymbol{\mathcal{L}} = \boldsymbol{\mathcal{D}} - \boldsymbol{\mathcal{A}}$. An edge sequence representing a straight path from node i to node j is written as $\{(\boldsymbol{\mathcal{V}_i}, \boldsymbol{\mathcal{V}_k}), (\boldsymbol{\mathcal{V}_k}, \boldsymbol{\mathcal{V}_l}), \cdots, (\boldsymbol{\mathcal{V}_m}, \boldsymbol{\mathcal{V}_j})\}$. Digraph that has a Minimum Spanning Tree is a digraph having a Minimum Spanning Tree if every node in the communication network has a direct path between every other node and its root node.

Supposing agents in the network are connected using a communication graph. The control method requires a communication graph with adjacency, degree, and Laplacian matrices [27], [28]. The adjacency matrix, $\boldsymbol{\mathcal{A}(\mathcal{G})}$, is formulated as:

$$\boldsymbol{\mathcal{A}(\mathcal{G})} = [\boldsymbol{a_{ij}}] \begin{cases} \boldsymbol{1} & \boldsymbol{\mathcal{V}_i \mathcal{V}_j \in N} \\ \boldsymbol{0} & \boldsymbol{otherwise} \end{cases} \tag{2}$$

where $\boldsymbol{a_{ij}}$ is the communication weight (a design parameter), and N, the neighboring nodes. The minimum spanning tree (MST) ensures the optimal path for information exchange among the agents. The communication graph with MST must be constructed before communication can start. The MST must also be updated when a new agent joins the system [29]. The communication graph should at least have a MST, be directional and carry the minimum redundancy. A degree of a vertex is denoted $\boldsymbol{d(\mathcal{V}_i)}$, in undirected graph $\boldsymbol{\mathcal{G}_r}$, this is given by the number of adjacent vertices to $\boldsymbol{v_i}$ in the graph. The degree matrix, $\boldsymbol{\mathcal{D}(\mathcal{G})}$, is a diagonal matrix where the diagonal elements is represented by the degree of each vertex [30]. The degree matrix can be represented as shown in Equation 3:

$$\boldsymbol{\mathcal{D}(\mathcal{G})} = \begin{bmatrix} \boldsymbol{d(\mathcal{V}_1)} & \boldsymbol{0} & \cdots & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{d(\mathcal{V}_2)} & \cdots & \boldsymbol{0} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{0} & \boldsymbol{0} & \cdots & \boldsymbol{d(\mathcal{V}_n)} \end{bmatrix} \tag{3}$$

The graph Laplacian matrix, $\boldsymbol{\mathcal{L}(\mathcal{G})}$, associated with the degree and adjacency matrices, is given as

$$\boldsymbol{\mathcal{L}(\mathcal{G})} = \boldsymbol{\mathcal{D}(\mathcal{G})} - \boldsymbol{\mathcal{A}(\mathcal{G})} \tag{4}$$

The uniqueness of the Laplacian matrix, $\boldsymbol{\mathcal{L}(\mathcal{G})}$ is described in the following: $\boldsymbol{\mathcal{L}}$ is positive semi-definite i.e., it takes up a real matrix $\boldsymbol{\mathcal{A}}$ denoted as positive definite if and only if the eigenvalues of $\boldsymbol{\mathcal{A} + \mathcal{A}^T}$ are non-negative. The smallest eigenvalue of $\boldsymbol{\mathcal{L}}$ is zero, other eigenvalues are non-negative and real $(\boldsymbol{0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n})$. And finally, the addition of individual Laplacian matrix row equals to zero.

2.2. Microgrid Control Structure

Standalone inverter-based microgrid considered in this paper consists of four distributed generating (DG) units with Photovoltaic (PV) solar energy source. For the i_{th} DG, the primary controller (which comprises the conventional power droop, voltage and current controllers in a loop), as well as the secondary control unit are as shown in Fig. 1. The d-q reference frame transforms of the original abc waveform is used, where d-axis and q-axis of each DG rotates with the common reference frequency. In the primary control loop, the control procedure implemented during operation is as follows: the power droop control loop is fed with the nominal set points ω_{ni} and e_{ni}. The power droop controller generates the reference values, ω_i^* and v_{odi}^*, which serves as the inputs of the voltage controller, the voltage loop generates the current reference, i_{ldi}^* and i_{lqi}^*, for the current control loop. Then, in the current controller, the current errors are determined and the voltage reference signals v_{ldi}^* and v_{lqi}^* generated are used to regulate the outputs of the inverter via Space Vector Pulse Wave Modulator (SVPWM).

To mitigate the deviations in frequency as a result of the primary control procedure, the secondary control mechanism is implemented in a distributed manner to generate new nominal set points for the primary controller of the DGs. The multi-agent control scheme is used to design the secondary controller using the information shared among the neighboring DGs to update ω_i^* and v_{odi}^* in each primary control process, and thus, restore the terminal output ω_{ni} and e_{ni} , to the desired reference values.

3. Multi-Agent Communication

The communication scenario in the microgrid system is depicted in Fig. 2. The four communicating DGs shown are connected together. When information is being shared among the DGs, impairments is introduced along with the shared information.

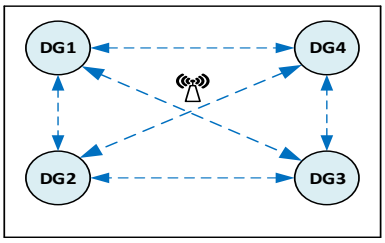


Figure 2: Graph topology for the communication network with full connectivity.

The interactive matrix (adjacency matrix), a_{ij1}, that describes the communication between the DGs in the microgrid is displayed in (5). Since the graph in Fig. 2 is fully linked and undirected, all DGs are able to transmit information in both directions. When the weight of each edge is 1, it indicates strong connection.

$$a_{ij1} = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \tag{5}$$

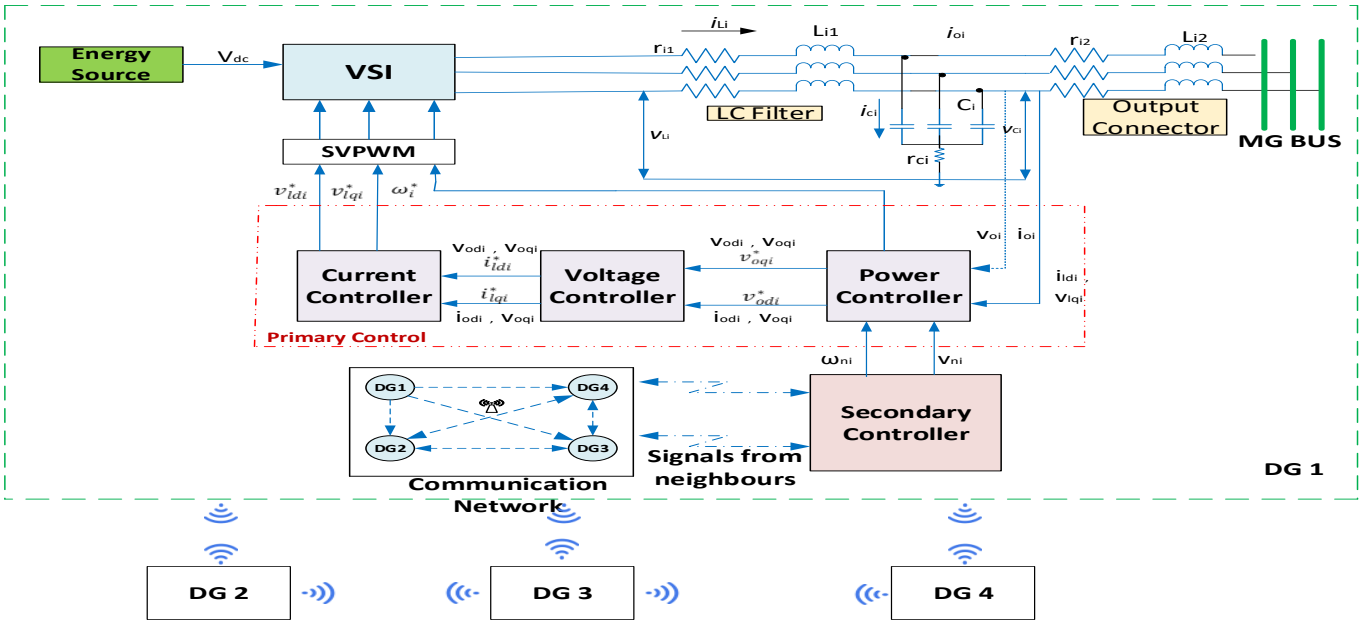


Figure 1: Distributed generator control framework.

Then using (6) and (7) degree and Laplacian matrices are derived.

$$\mathcal{D}(\mathcal{G}) = \begin{bmatrix} d(\mathcal{V}_1) & 0 & \cdots & 0 \\ 0 & d(\mathcal{V}_2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d(\mathcal{V}_n) \end{bmatrix} \quad (6)$$

where a degree of a vertex is denoted $d(\mathcal{V}_i)$, The graph Laplacian matrix, $\mathcal{L}(\mathcal{G})$, associated with the degree and adjacency matrices, is defined as

$$\mathcal{L}(\mathcal{G}) = \mathcal{D}(\mathcal{G}) - \mathcal{A}(\mathcal{G}) \quad (7)$$

3.1. Algorithm for DGs Communication

Assumptions: All the multi-agent units are within the communication range of each other and communication between agents are continuous. Also, all units are synchronized to the time of communication. Packet level synchronization precision is required.

Let τ be the time instant. At ith time instant, DG_j is transmitting while DG_i is receiving, provided $j \epsilon N_i$. The set of neighbours of node v_j is denoted by (1) where ε, represent the edge connecting the agents nodes together; v_i and v_j are the nodes at DG_i and DG_j respectively. The condition implies that the communication is achievable using communication graph with minimum spanning tree. Parameters to be transmitted is measured by each of the DG at each instant of time, the parameters are ω_i, and P_i, that is, frequency, and active power.

The algorithm for consensus of multi-agent system is described as follows: Considering DG_i, at ith time instant, ω_i, and P_i, are measured and denoted by $x_i(\tau)$, the trendy local estimate of the global average of the measured parameters calculated at DG_i at time instant τ is denoted by $h_i(\tau)$. The signal transmitted by DG_j at ith time instant is given as follows:

$$T(\tau) = h_j(\tau); \;\; j \epsilon N_i \quad (8)$$

DG_j transmits current state value which all neighboring DGs received. Upon the reception of the transmitted value, the neighboring DGs in the system update local estimate as follows:

$$h_i(\tau) = x_i(\tau) + \delta x_i(\tau), \qquad i = 0, 1, \ldots . N-1 \quad (9)$$

δx_i is the local control signal estimated based on the last received transmitted signal $h_j(\tau)$.

$$h_j(\tau) = h_j(k) \quad (10)$$

k denotes the time instant at which the last successful transmitted signal has been received by agent DG_i.

3.2. Communication Link Impairments

3.2.1. Communication Link Noise

Considering two agents (nodes), agent v_i, the receiver and agent v_j, the transmitter, communicating wirelessly. The following steps are taken to arrive at the model for communication link noise used in the simulation: Signal transmission: Node v_j transmit a signal $x_j(t)$, with power S; wireless channel: The signal travels through the wireless channel, which introduces noise, $n(t)$. Received Signal: Node v_i receives a signal $r(t)$, which is the sum of the original signal and noise: $r(t) = x_j(t) + n(t)$. Using the linear consensus

protocol the expression for wireless communication link noise can be given as:

$$r_i(t) = \sum_{j \in N_i} a_{ij}\left[x_i(t) - x_j(t)\right] + n(t) \quad (11)$$

where $n(t)$ is the noise which represents Additive White Gaussian Noise (AWGN).

3.2.2. Communication Link Time Delay

Considering a network with a fixed topology $\mathcal{G} = (v, \varepsilon, A)$ for sets of node where the information of the node state v_j is transmitted to a neighboring node v_i through a communication link e_{ji}. Applying the following linear protocol [31]:

$$u_i(t) = \sum_{u_j \in N_i} a_{ij}\left[x_j\left(t - \tau_{ji}\right) - x_i\left(t - \tau_{ji}\right)\right] \quad (12)$$

A consensus gain, $-K_{c,i}$, is introduced to (12), the state of the node is then expressed as:

$$\dot{x}_i(t) = -K_{c,i} \sum_{u_j \in N_i} a_{ij}\left[x_j\left(t - \tau_{ji}\right) - x_i\left(t - \tau_{ji}\right)\right] \quad (13)$$

Assuming that $\psi_i \in \mathcal{R}$ denotes state variables (frequency or active power) of ith agent, the information sharing process with time delay τ_{ij} can be expressed as:

$$\psi_i\left(t + \tau_{ij}\right) = \psi_i(t) + \sum_{j \in N_i} a_{ij}\psi_j(t) \quad (14)$$

where $i = 1, 2, \ldots . n; j = 1, 2, \ldots . n; \psi_i(t)$ and $\psi_i\left(t + \tau_{ij}\right)$ are the shared information by ith DG at iteration t and $t + \tau_{ij}$ respectively. $\psi_j(t)$ is the information shared by jth DG. a_{ij} is the adjacency matrix which represent the coefficient for information exchange between agent v_i and v_j. The global information sharing process of the microgrid can be expressed as:

$$\begin{cases} \psi\left(t + \tau_{ij}\right) = \mathcal{A}_{ij} \times \mu(t) \\ \tau_{ij} = \dfrac{n_{bit} \times n_p}{\mathcal{V}_{com}} + \left(\dfrac{d}{s}\right) \end{cases} \quad (15)$$

where $\mu(t)$ is the information sharing matrix, $\mathcal{A}_{ij}$ is the updating matrix defined in relation to the communication topology, τ_{ij} is the time delay of information sharing in one iteration process, it also represents the communication time delay. n_p is the elements in the information sharing matrix, $\mathcal{V}_{com}$ is the data communication speed in terms of bits per second, d is the distance separating the DGs or communicating agents and s is the speed of propagation in the medium between two agents. For a strong and balanced connection between agent v_i and v_j on the communication graph topology, $\mathcal{A}_{ij}$ satisfies the following:

$$\begin{cases} \sum_i a_{ij} = 1 \\ \sum_j a_{ij} = 1 \end{cases} \quad (16)$$

4. Consensus-Based Secondary Frequency Control

The following assumptions were considered in the development of the consensus-based control system: i. each of the Distributed Generating (DG) unit is regarded as an "agent" which is capable of sharing information with other DGs in its neighbourhood. ii. each agent can also take control decisions based on the information received from the neighbours. iii. all participating agents aim to reach a consensus with the values being shared, and the reference value is the predetermined consensus value.

4.1. Frequency Synchronization

Given the frequency droop expressed by Equation 17

$$\omega_i^* = \omega_{ni} - m_{pi}P_i \quad (17)$$

where ω_{ni}, is the rated system frequency, m_{pi} is the droop coefficient, and P_i is the measured active power. ω_{ni}, is the actual frequency from DGi. Using the shared variable, ω_{ni}, and P_i, the objective is to design $\delta\omega_i$, and δp_i, the distributed frequency and active power controller respectively, that will satisfy:

$$\lim_{t \to \infty} \omega_{ni}(t) = \omega_{MG} \quad i = 1, 2, \cdots, N \quad (18)$$

That is, to restore the frequency of each DG to the reference value, and as a result reach a consensus value for the microgrid optimal operation.

The second objective is to ensure optimal active power sharing among the DGs, that is;

$$m_1P_1 = m_2P_2 = \cdots = m_NP_N, as\ t \to \infty \quad (19)$$

These objectives are to be achieved despite the effects of communication impairments considered in this paper. Using the consensus protocol and incorporating the outputs of neighbouring DGs, the frequency controller $\delta\omega_i$ is written as:

$$\delta\omega_i = \int \left[C_1 \sum_{j \in N_i} a_{ij}\left(\omega_j\left(t - \tau_{ij}\right) - \omega_i(t)\right) - g_i\left(\omega_i(t) - \omega_{ref}\right) \right] dt \quad (20)$$

where C_1 is a positive constant representing the coupling strength of the communication network, a_{ij} is the weighted adjacency matrix of the frequency, g_i is the pinning gain, which is 1 when the node is connected to the reference value or 0 otherwise.

4.2. Active Power Sharing

Using the consensus-based algorithm, the objective of equal active power sharing could be achieved. The power sharing algorithm could be designed by incorporating relative active power among neighbouring DGs as follows:

$$\delta P_i = \sum_{j \epsilon N_i} C_3 a_{ij} \left[m_j P_j \left(t - \tau_{ij} \right) - m_i P_i \right] \Big/ m_i \quad (21)$$

For all $i = 1,2,. \ . \ . \ N$, where C_3 is a positive constant representing the coupling strength of the communication network, m_j and m_i are the constant droop coefficients of nodes v_j and v_i respectively.

The eventual control signal is given by:

$$u_i^{\omega} = \delta\omega_i + \delta P_i \quad (22)$$

The general framework for the multi-agent control system is given in Fig. 3. The multi-agent control inputs have two parts, the part shown in (20), is the summation of the tracking errors in frequency based on the information from the neighbours. This component is employed for the correction of frequency deviation. In the second part of (21), the summation is of the local active power tracking error with neighbours. For the active power sharing, this aspect is used.

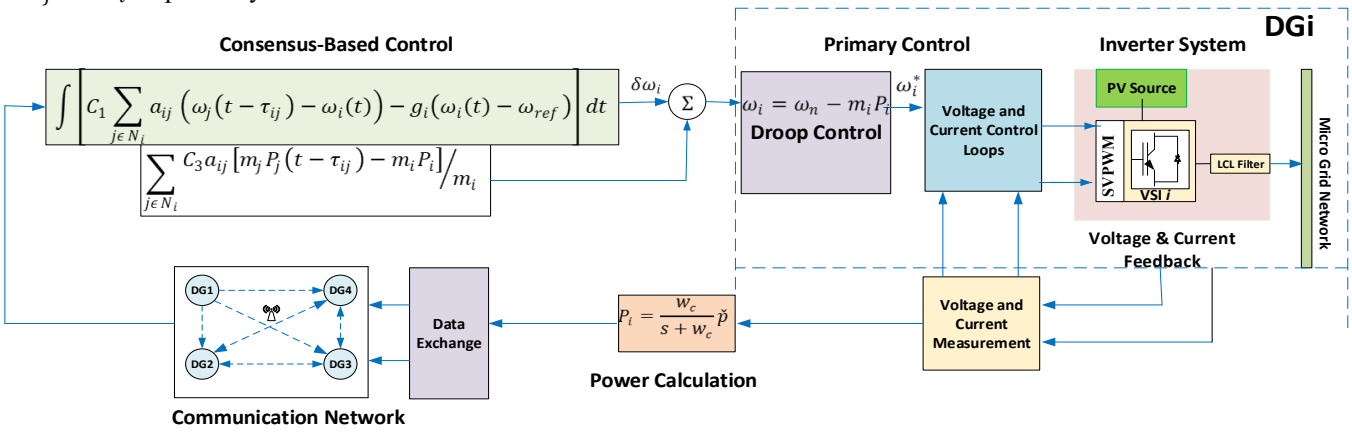


Figure 3: Framework for the consensus-based secondary frequency control scheme.

4. Simulation Study

A four-inverter microgrid test system, shown in Fig. 4, is prototyped in the MATLAB/SIMULINK software environment. The rated voltage and frequency are 240 V and 60 Hz respectively. Filters are connected at the inverters' outputs to reduce switching harmonics. Parameters used for simulations are given in Table 1. More DGs mean more agents participating in frequency regulation, which could increase the complexity of control coordination. Increased data exchange and latency can affect performance. However, more DGs can enhance frequency support and fault tolerance. In well-designed distributed systems, additional DGs can actually improve robustness if control effort is well distributed. While the proposed method demonstrates effectiveness in a four-DG microgrid, its scalability to larger systems necessitates further investigation. Key considerations include communication delays, distributed control coordination, and robustness under tighter operational constraints. Future work will focus on evaluating the method's performance under dynamic topologies and fault scenarios to ensure viability in more complex microgrid configurations.

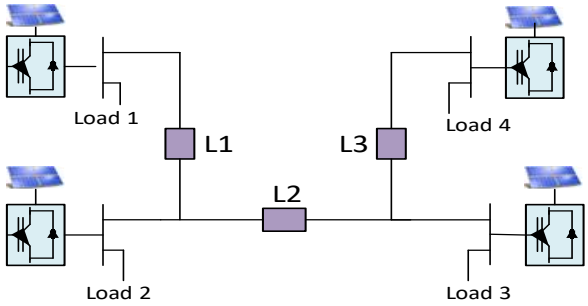


Figure 4: Single-line diagram of the microgrid test system.

Table 1. Parameters for the microgrid test system.

DG1 & DG2		DG3 & DG4		Lines	R(Ω)	L(mH)
m_p	9.4e-5	m_p	12.5e-5	1	0.23	0.318
n_q	1.3e-3	n_q	1.5e-3	2	0.35	0.184
R_c	0.03Ω	R_c	0.03Ω	3	0.23	0.318
L_c	0.35mH	L_c	0.35mH			
R_f	0.1Ω	R_f	0.1Ω			
L_f	1.35mH	L_f	1.35mH	**Loads**	**(kW)**	**L(mH)**
C_f	50μF	C_f	50μF	1	14	0.5
K_{pv}	0.1	K_{pv}	0.05	2	13	0.5
K_{iv}	420	K_{iv}	390	3	11	0.5
K_{pc}	15	K_{pc}	10.5	4	10	0.5
K_{ic}	20,000	K_{ic}	16,000			

5. Results and Discussion

Results of the simulation work carried out are discussed under this section.

5.1. Performance Under Load Perturbation Without Communication Impairments

In this subsection, the performance of the proposed secondary control scheme is under the following simulation stages: Stage 1 ($0.0s \leq t < 0.1s$): only primary droop control scheme is active. Stage 2 ($0.1s \leq t < 0.3s$): At t = 0.1s, the load on each of the DGs was doubled by adding a load of 1 kW, and L of 0.5mH from DG 1 to DG 4. Stage 3 ($t \geq 0.3s$): At t = 0.3s, the proposed consensus-based secondary controller was activated and remained active thereafter. The frequency response of the system is shown in Fig. 5, showing the frequency fluctuations of each DGs with time. It can be observed that the primary droop controller was able to keep the frequency fluctuations within an acceptable limit. According to IEEE standards; IEEE Std 1547a-2014 and IEEE Std 1547-2018, a frequency deviation of 0.5Hz is the acceptable limit of maximum frequency deviation. Also, the limit for over frequency is 0.8% while the limit for under frequency is 1%. For this simulation case of rated frequency of 60Hz, the values correspond to 0.48Hz for over frequency and 0.6Hz for under frequency. At the instance of load perturbation, some frequency fluctuations can be observed but the fluctuations were kept within the acceptable limit. Upon activation of the consensus-based secondary controller, some frequency fluctuations also occurred but of much reduced deviation. Before the activation of the secondary controller, only the droop controller was in operation, and the system took some time to reach frequency consensus. However, the performance of the consensus-based secondary control scheme indicates that frequency was restored to the operating value of 60Hz faster and subsequently maintained.

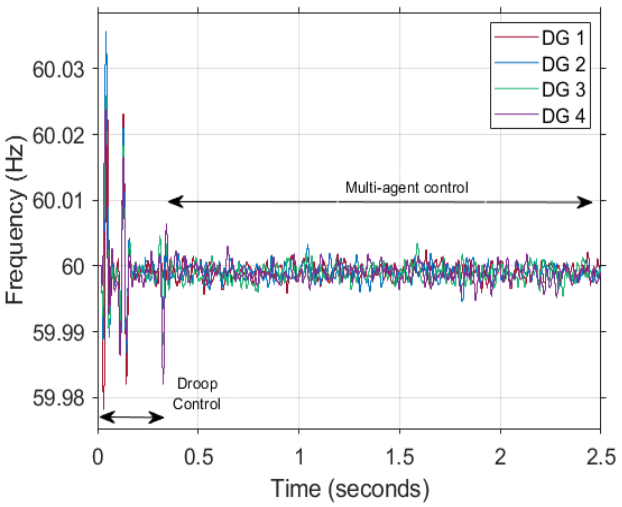
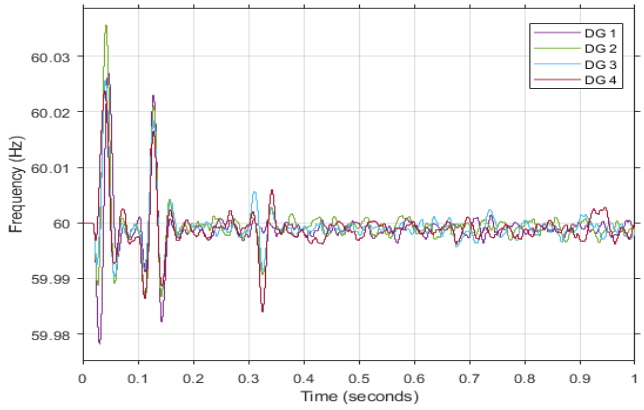


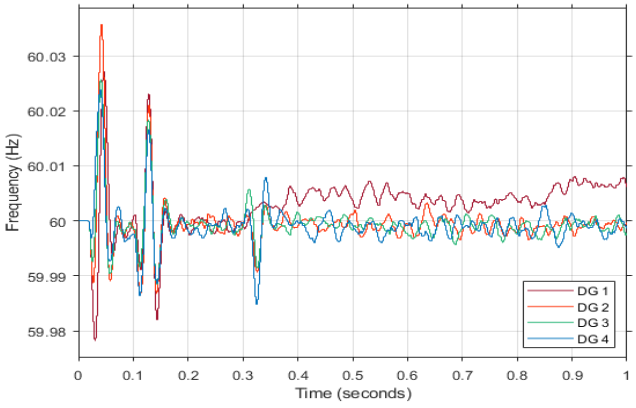
Figure 5: Tracking frequency restoration and synchronization of the proposed control scheme.

5.2 Performance Under Different Noise Levels

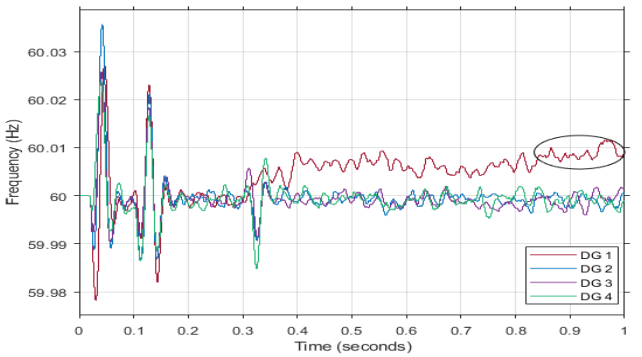
The communication between DG1 and other DGs in the microgrid was corrupt with communication noise, time delay and link failure to evaluate performance of the controller under communication conditions. The secondary frequency controller performance was then evaluated. The consensus-based controller frequency restoration performance under noise impairment, with the following noise variance, σ^2 of 0, 0.1, 0.2, 0.3, 0.4 and 0.5 were considered. The waveforms shown in Fig. 6 show that the DGs' frequencies were synchronized under acceptable limits for different noise levels. For $\sigma^2 = 0$ (Fig. 6(a)), the DGs' frequencies quickly reached consensus after the activation of secondary control just like that in Fig. 5. For σ^2 above 0, there are some deviations in the frequency of DG 1 occasioned by the noise impairment between it and the other DGs as shown in Fig. 6(b)-(f), but the distortion was kept within acceptable limits. The higher the noise variance, the more the distortion. When σ^2 = 0.1, 0.2, 0.3, 0.4 and 0.5, the frequency distortions after activation of the secondary controller, were less than 0.01Hz, 0.015Hz, 0.02Hz, 0.03Hz and 0.05Hz, respectively. However, for increased noise level $\sigma^2 = 0.4$ and 0.5 (Fig. 6 (e) and (f)), however, the distortion grows noticeably but is still acceptable. It can be seen that increasing the noise level degrades frequency of synchronization in these scenarios.



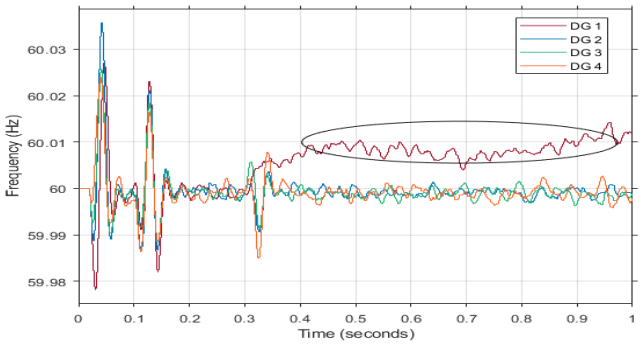
(a)



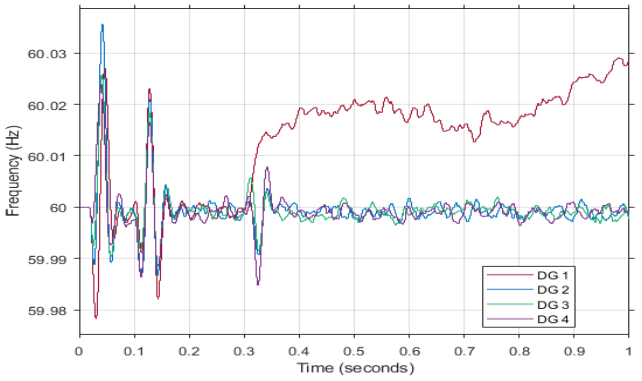
(b)



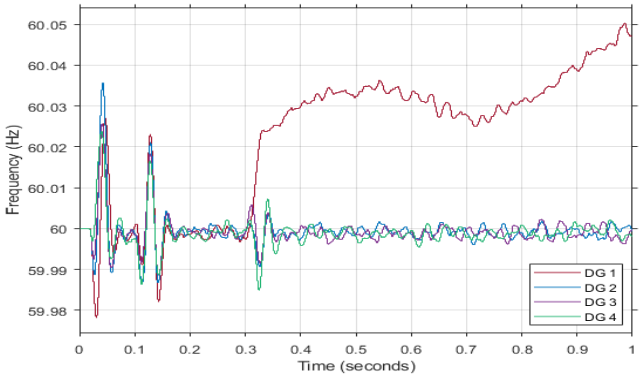
(c)



(d)



(e)

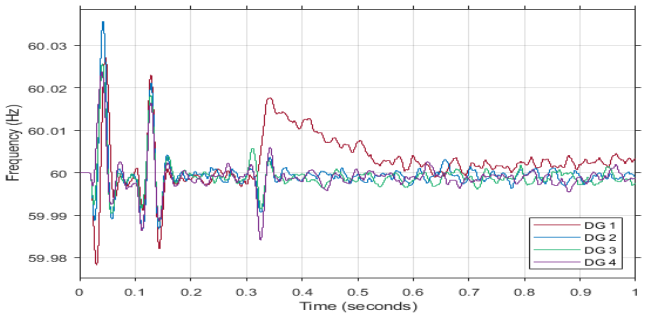


(f)

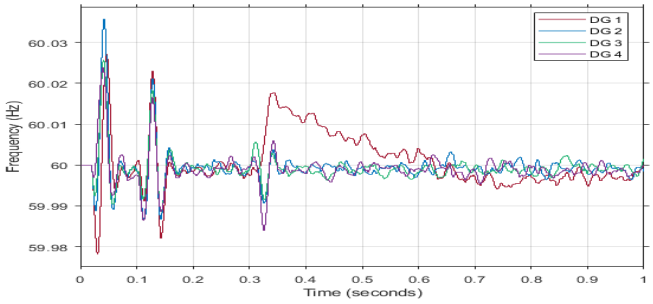
Figure 6: Tracking multi-agent controller for frequency regulation performance (a) $\sigma^2 = 0$; (b) $\sigma^2 = 0.1$; (c) $\sigma^2 = 0.2$; (d) $\sigma^2 = 0.3$; (e) $\sigma^2 = 0.4$; (f) $\sigma^2 = 0.5$.

5.3 Performance with Communication Delay

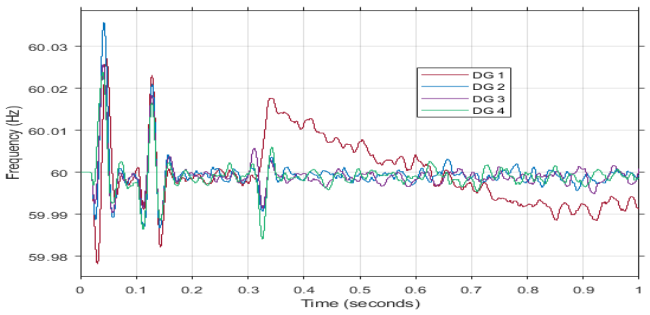
Frequency performance under communication delay impairment was investigated for the following latency between DG 1 and other DGs: $\tau_{t1} = 0.6s$; $\tau_{t2} = 0.7s$, $\tau_{t3} = 0.8s$ and $\tau_{t4} = 1.0s$. The waveforms shown reveal that the DGs' frequencies were synchronized within acceptable limits for different latency. For τ less than 1.0s, the consensus-based control scheme was able to synchronize the delayed DG 1 frequency with the remaining DGs in the system. The distortion in DG 1 which was occasioned by the communication delay impairment was kept within acceptable limits. Since secondary frequency control was activated at t = 0.3s, note that for τ of 0.6s and 0.7s, DG 1 frequency reached consensus with others by 0.7s and subsequently maintained. For τ of 0.8s (Fig. 7(c)), DG 1 frequency first reaches consensus with others at 0.7s but was not maintained. It will reach consensus with others after 1.0s. However, for increased delay at $\tau = 1.0$s (Fig. 7(d)), the deviation in the frequency of DG 1 increases noticeably but it is still within acceptable limits and will reach consensus with others well after 1.0s. This shows that increased communication delay could result in system instability.



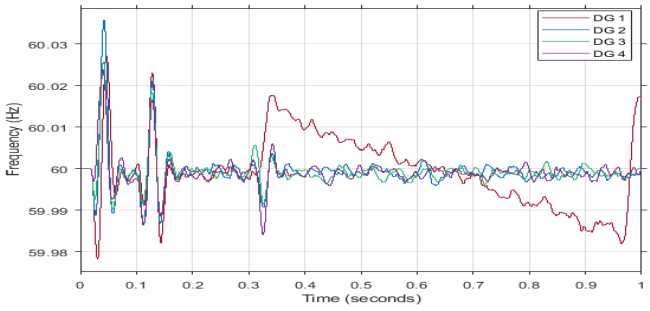
(a)



(b)



(c)

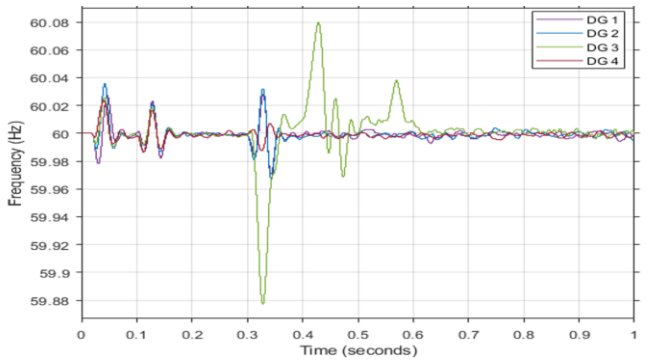


(d)

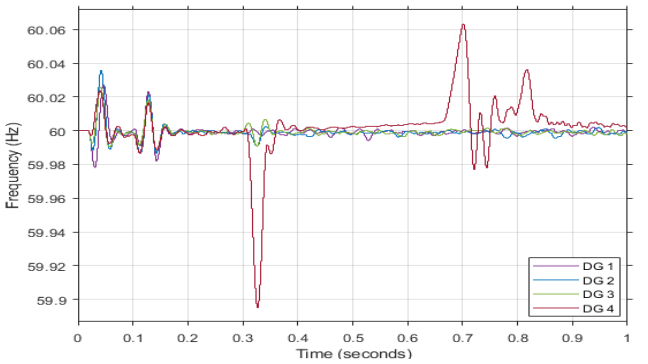
Figure 7: Tracking frequency performance of the multi-agent control scheme under communication delay (a) $\tau_{t1} = 0.6s$; (b) $\tau_{t2} = 0.7s$; (c) $\tau_{t3} = 0.8s$; and (d) $\tau_{t4} = 1.0s$.

5.3 Performance with Link Failure

Fig. 8 illustrates the performance of the multi-agent controller with link failure. In Fig. 8(a), the link between DG 3 and DG 2 failed, causing frequency of DG 3 to fluctuate between 59.88 Hz and 60.08 Hz after activation of the secondary controller. It takes about 0.3 seconds to reach consensus with others i.e. it reaches consensus with others at approximately 0.62 seconds. In Fig. 8(b), there was link failure between DG 4 and DG 3, and the frequency of DG 4 fluctuates between 59.895 Hz and 60.06 Hz after activation of the secondary controller. Consensus was reached at approximately 1.0 seconds.



(a)



(b)

Figure 8: Tracking frequency synchronization performance of the multi-agent control scheme under communication failure (a) link failure in DG 3; (b) link failure in DG 4.

6. Comparison with Existing Methods

The performance of the proposed consensus-based control method shown in Table 2, was evaluated and compared with the approaches presented in references [1] and [2], under various communication impairment scenarios. The test conditions included a communication delay (τ) of 1.0 seconds, a link noise variance (σ^2) of 0.3, and a communication link failure between DG1 and DG4. Under the influence of communication delay and noise, the proposed method outperformed the methods in [1] and [2], achieving superior frequency regulation with over-frequency and under-frequency deviations limited to 0.03 Hz and 0.02 Hz, respectively, and a convergence time of 1 second. Additionally, when subjected to noise with a variance of 0.3, the developed method demonstrated improved stability and robustness compared to the benchmark methods. However, in the case of a communication link failure between DG1 and DG4, the method in [32] exhibited greater resilience, maintaining more stable system behavior. Overall, the proposed consensus-based secondary control strategy demonstrated commendable performance in mitigating communication impairments, including delay, noise, and partial link failures, within inverter-based islanded microgrids.

Table 2. Comparison with existing methods.

Impairment	Method	Frequency Deviation		Time to Reach Convergence (Sec.)
		Over Freq. (%)	Under Freq. (%)	
τ =1.0 (sec.)	Ref. [1]	0.1	0.6	5
	Ref. [2]	0.12	0.01	2
	Proposed Method	0.03	0.02	1
$\sigma^2 = 0.3$	Ref. [1]	-	-	-
	Ref. [2]	0.2	0.18	1.8
	Proposed Method	0.01	0	0.4
Link failure bet. DG1 and DG4	Ref. [1]	0.2	0	1
	Ref. [2]	0	0	0
	Proposed Method	0.02	0.02	0.6

7. Conclusion

A proposed method of consensus-based secondary frequency control by error tracking of frequency and active power to ensure frequency synchronization, and to mitigate frequency deviation in autonomous microgrids with communication link impairments has been presented and simulated. The study findings highlight the adequacy of secondary control in effectively managing frequency deviation and synchronization within microgrids. Although the primary droop controller keeps the maximum frequency deviation within allowable limits, it is weak in regulating the frequency back to its initial value. However, the proposed consensus-based secondary control scheme successfully regulates the frequency to its reference value within appreciable time despite impairments such as noise, link delay and link failure in the communication

links of the DGs in the MG, thereby ensuring minimal frequency distortion within acceptable limits. Too long communication delays could result in system instability and should therefore be prevented in the communication network. The results indicate that the proposed method effectively addresses communication impairment problems in microgrid operation, thus ensuring system resilience and enhancing proper power sharing among the generators.

Funding: This research received no external funding.

Ethical Statement: This research adheres to the ethical guidelines established by the Committee on Publication Ethics (COPE). All procedures and methodologies used in this study comply with COPE standards, ensuring transparency, integrity, and respect for all participants involved.

Conflicts of Interest: The authors declare no conflict of interest.

References

[1] G. Chen and Z. Guo, "Distributed secondary and optimal active power sharing control for islanded microgrids with communication delays," *IEEE Trans. Smart Grid*, vol. 10, no. 2, pp. 2002–2014, 2019, doi: 10.1109/TSG.2017.2785811.

[2] S. A. Olayanju, F. M. Dahunsi, and A. A. Ponnle, "Evaluation of Cross-Coupling Decoupling in Islanded Microgrid Control with Inductor Current Feed-Forward Technique," *EECCIS*, vol. 18, no. 3, pp. 107–116, 2024.

[3] K. Ahmed, M. Seyedmahmoudian, S. Mekhilef, N. M. Mubarak, and A. Stojcevski, "A Review on Primary and Secondary Controls of Inverter-interfaced Microgrid," *Journal of Modern Power Systems and Clean Energy*, vol. 9, no. 5. pp. 969–985, 2021. doi: 10.35833/MPCE.2020.000068.

[4] M. I. Habib, "Cybersecurity for Smart Inverters : State-of-the-Art Review," *Pakistan J. Eng. Technol.*, vol. 7, no. 4, pp. 151–158, 2024.

[5] S. Shrivastava, B. Subudhi, and S. Das, "Noise-resilient voltage and frequency synchronisation of an autonomous microgrid," *IET Gener. Transm. Distrib.*, vol. 13, no. 2, pp. 189–200, 2019, doi: 10.1049/iet-gtd.2018.6409.

[6] O. F. Rodriguez-Martinez, F. Andrade, C. A. Vega-Penagos, and A. C. Luna, "A Review of Distributed Secondary Control Architectures in Islanded-Inverter-Based Microgrids," *Energies*, vol. 16, no. 2, 2023, doi: 10.3390/en16020878.

[7] X. Lu, X. Yu, J. Lai, Y. Wang, and J. M. Guerrero, "A Novel Distributed Secondary Coordination Control Approach for Islanded Microgrids," 2018.

[8] V. Gurugubelli, A. Ghosh, and A. K. Panda, "Parallel inverter control using different conventional control methods and an improved virtual oscillator control method in a standalone microgrid," *Prot. Control Mod. Power Syst.*, vol. 7, no. 1, 2022, doi: 10.1186/s41601-022-00248-9.

[9] A. Afshari, M. Karrari, H. R. Baghaee, and G. B. Gharehpetian, "Resilient cooperative control of AC microgrids considering relative state- dependent noises and communication time- delays," *IET Renew. Power Gener.*, vol. 14, pp. 1321–1331, 2020, doi: 10.1049/iet-rpg.2019.1180.

[10] F. Tariq and T. Mahmood, "Modeling and Analysis of Voltage Source based Battery Energy Storage System in Microgrid to Improve Power Quality," *Pakistan J. Eng. Technol.*, vol. 4, no. 4, pp. 29–34, 2021.

[11] W. Li, M. Zhang, and Y. Deng, "Consensus-Based Distributed Secondary Frequency Control Method for AC Microgrid Using ADRC Technique," *Energies*, vol. 15, no. 9, 2022, doi: 10.3390/en15093184.

[12] S. A. Olayanju, F. M. Dahunsi, and A. A. Ponnle, "Mitigation strategies for communication networks induced impairments in autonomous microgrids control : A review," *AIMS Electron. Electr. Eng.*, vol. 5, no. 4, pp. 342–375, 2021, doi: 10.3934/electreng.2021018.

[13] Z. Wang, S. Member, W. Wu, and S. Member, "A Distributed Control Method with Minimum Generation Cost for DC Microgrids," vol. 8969, no. c, pp. 1–9, 2016, doi: 10.1109/TEC.2016.2584649.

[14] N. M. Dehkordi, H. R. Baghaee, N. Sadati, and J. M. Guerrero, "Distributed Noise-Resilient Secondary Voltage and Frequency Control for Islanded Microgrids," *IEEE Trans. Smart Grid*, vol. 10, no. 4, pp. 3780–3790, 2019, doi: 10.1109/TSG.2018.2834951.

[15] H. Li, H. Hui, and H. Zhang, "Consensus-Based Energy Management of Microgrid with Random Packet Drops," *IEEE Trans. Smart Grid*, vol. 14, no. 5, pp. 3600–3613, 2023, doi: 10.1109/TSG.2023.3241653.

[16] N. Mosaad, O. Abdel-Rahim, T. F. Megahed, W. Rohouma, T. Asano, and S. M. Abdelkader, "An enhanced consensus-based distributed secondary control for voltage regulation and proper current sharing in a DC islanded microgrid," *Front. Energy Res.*, vol. 11, no. November, pp. 1–19, 2023, doi: 10.3389/fenrg.2023.1277198.

[17] C. Zhao, W. Sun, J. Wang, and Z. Fang, "Distributed robust secondary voltage control for islanded microgrid with nonuniform time delays," *Electr. Eng.*, vol. 103, no. 6, pp. 2625–2635, 2021, doi: 10.1007/s00202-021-01251-3.

[18] C. Peng, S. Member, H. Sun, M. Yang, and Y. Wang, "A Survey on Security Communication and Control for Smart Grids Under Malicious Cyber Attacks," *IEEE Trans. Syst. Man, Cybern. Syst.*, vol. PP, pp. 1–16, 2018, doi: 10.1109/TSMC.2018.2884952.

[19] C. Zhao, W. Sun, J. Wang, Q. Li, D. Mu, and X. Xu, "Distributed Cooperative Secondary Control for Islanded Microgrid With Markov Time-Varying Delays," *IEEE Trans. Energy Convers.*, vol. 34, no. 4, pp. 2235–2247, 2019.

[20] X. Lu, X. Yu, J. Lai, J. M. Guerrero, and H. Zhou, "Distributed Secondary Voltage and Frequency Control for Islanded Microgrids With Uncertain Communication Links," *IEEE Trans. Ind. Informatics*, vol. 13, no. 2, pp. 448–460, 2017, doi: 10.1109/TII.2016.2603844.

[21] C. Zhao, W. Sun, J. Wang, Q. Li, D. Mu, and X. Xu, "Distributed cooperative secondary networked optimal control with packet loss for islanded microgrid," vol. 13, no. i, pp. 4733–4740, 2019, doi: 10.1049/iet-gtd.2019.0472.

[22] M. Yaribeygi, Z. Karami, Q. Shafiee, and H. Bevrani, "Distributed Hybrid Model Predictive Secondary Control of DC Microgrids with Random Communication Disorders," *Mordern power Syst. Clean Energy*, vol. 12, no. 3, pp. 925–935, 2024, doi: 10.35833/MPCE.2023.000417.

[23] P. Cai, C. Wen, and C. Song, "Consensus-based Secondary Frequency Control for Islanded Microgrid with Communication Delays," *2018 Int. Conf. Control. Autom. Inf. Sci.*, no. Iccais, pp. 107–112, 2018.

[24] C. Dou, B. Zhang, D. Yue, Z. Zhang, S. Xu, and T. Hayat, "A novel hierarchical control strategy combined with sliding mode control and consensus control for islanded micro-grid," *IET Renew. Power Gener.*, vol. 12, no. 9, pp. 1012–1024, 2018, doi: 10.1049/iet-rpg.2017.0754.

[25] R. Olfati-Saber and R. M. Murray, "Consensus problems in networks of agents with switching topology and time-delays," *IEEE Trans. Automat. Contr.*, vol. 49, no. 9, pp. 1520–1533, 2004, doi: 10.1109/TAC.2004.834113.

[26] H. Zhang, S. Kim, Q. Sun, and J. Zhou, "Distributed Adaptive Virtual Impedance Control for Accurate Reactive Power Sharing Based on Consensus Control in Microgrids," *IEEE Trans. Smart Grid*, vol. 8, no. 4, pp. 1749–1761, 2017, doi: 10.1109/TSG.2015.2506760.

[27] G. Lou, W. Gu, X. Lu, Y. Xu, and H. Hong, "Distributed Secondary Voltage Control in Islanded Microgrids with Consideration of Communication Network and Time Delays," *IEEE Trans. Smart Grid*, vol. 11, no. 5, pp. 3702–3715, 2020, doi: 10.1109/TSG.2020.2979503.

[28] V. Nasirian, S. Moayedi, A. Davoudi, and F. L. Lewis, "Distributed cooperative control of dc microgrids," *IEEE Trans. Power Electron.*, vol. 30, no. 4, pp. 2288–2303, 2015, doi: 10.1109/TPEL.2014.2324579.

[29] V. Lakshminarayanan, V. C. Sekar, and K. Rajashekara, "Novel communication architecture for Multi-Agent Systems in autonomous Microgrid," in *IEEE Power and Energy Society General Meeting*, 2016, vol. 2016-Novem. doi: 10.1109/PESGM.2016.7741582.

[30] B. Ali, V. Nasirian, A. Davoudi, and F. L. Lewis, *Cooperative Synchronization in Distributed Microgrid Control*, vol. 1, no. April. 2017.

[31] O. S. Akinwale, D. F. Mojisola, and P. A. Adediran, "Consensus Issues in Multi-Agent-Based Distributed Control with Communication Link Impairments," *Niger. J. Technol. Dev.*, vol. 21, no. 1, pp. 85–93, 2024, doi: 10.4314/njtd.v21i1.2212.

[32] S. IEEE, *IEEE Standard for Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces Sponsored by the IEEE Standard for Interconnection and Interoperability of Distributed Energy Resources with Associate*. 2018.

This work is licensed under a Creative Commons Attribution 4.0 International License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.